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Thematic analysis with open-source generative AI and machine learning: a new method for inductive qualitative codebook development

Andrew Katz^{1✉}, Gabriella Coloyan Fleming¹ & Joyce B. Main²

This work aims to answer one central question: *to what extent can open-source generative text models be used in a workflow to approximate steps in thematic analysis in social science research?* To answer this question, we present the Generative AI-enabled Theme Organization and Structuring (GATOS) workflow, which uses open-source machine learning techniques, natural language processing tools, and generative text models to facilitate aspects of thematic analysis. To establish evidence of validity of the method, we present three case studies applying the GATOS workflow, leveraging these models and techniques to inductively create codebooks similar to traditional procedures using thematic analysis. We show that the GATOS workflow can identify themes in the text that were used to generate the original synthetic datasets. We conclude with a discussion of relevant considerations, the implications of this work for social science research, and the tradeoffs of using open-source generative text models to facilitate scalable qualitative data analysis.

¹Department of Engineering Education, Virginia Polytechnic Institute and State University, Blacksburg, VA, USA. ²School of Engineering Education, Purdue University, West Lafayette, IN, USA. ✉email: akatz4@vt.edu

Introduction

In social science research, one may want to answer questions about individuals' experiences, beliefs, and attitudes. To do so, the researcher can collect quantitative and/or qualitative data. Qualitative and quantitative methodologies have their own underlying epistemological assumptions and traditions (Salomon, 1991; Halfpenny, 1979). Qualitative data allow researchers to obtain nuanced, rich, descriptive insights. As Saldaña (2011) describes, qualitative data analysis (QDA) follows methods for meaning-making that humans have been using since our ancestors: constructing patterns and categories; identifying interactions and interrelationships; and using deductive, inductive, and abductive reasoning.

While there are many mathematically derived ways to analyze quantitative data, qualitative data have a different foundation and set of approaches for analysis. One form of qualitative data analysis involves a process called coding, the process of identifying themes, patterns, and insights in a corpus of text (Saldaña, 2011). This process is often done manually and can be time-consuming and labor-intensive. Consequently, it does not scale well to large volumes of text, which a researcher may want to do if trying to collect data that support inferential claims. For example, researchers may want to understand factors associated with different cultures of ethical behavior. Quantitative measures could provide some insights into this question, but qualitative responses to open-ended surveys collected from thousands of participants across dozens of organization types would provide a richer dataset. While qualitative research does not often aim for generalizability as a guiding purpose, some researchers may want to find a balance between the richness of qualitative data and the ability to make generalizable observations about populations of interest. Whereas a small group of researchers can analyze tens or hundreds of observations, thousands or tens of thousands present unique challenges for resources and consistency. These issues facing qualitative researchers in social research who want to work on a large scale are the central challenges motivating this paper. The purpose of this paper is to provide evidence toward the validity of a method that uses modern machine learning techniques, especially open-source large language models, to analyze large volumes of text and generate qualitative codebooks. We also aim to equip qualitative researchers interested in LLM-assisted QDA with foundational knowledge to conduct it.

Our approach for doing this validation work is simple: compare our new method and its resulting codebooks with the themes and sub-themes that were intentionally embedded into the synthetic data generated for this study. To do this, we simulate three datasets that imitate possible data collection and contexts in social science research. The first dataset is designed to mimic a study of teamwork dynamics, specifically teammate feedback, which is a common research topic in organizational behavior and management research (Wu et al., 2014; Potosky et al., 2022; Dominick et al., 1997; Donia et al., 2015). The second dataset is motivated by studies of organizational cultures of ethical behavior (Key, 1999; Roy et al., 2024; DeBode et al., 2013; Kaptein, 2011; Kuye et al., 2013). The third dataset is inspired by research on employee perspectives about returning to their offices after the COVID-19 pandemic (Gibson et al., 2023; Wang et al., 2021; Liu et al., 2020; Hou et al., 2021). We adopt this approach of using synthetic data as a set of complementary approaches intended to accumulate evidence to support the validation of the workflow we are presenting. In particular, the current paper presents results from using synthetic data so that we can intentionally control which themes appear in the data and variation in contexts (as listed above). Another complementary approach would involve comparing analyses with our workflow and human analyses of datasets collected from actual research studies. We are also

conducting that study but treat it as separate to provide the time and space to consider the results here using this synthetic data approach. Before describing the method and synthetic datasets, we briefly review the literature on qualitative coding and large language models in social science research.

Background

Qualitative codebooks and coding. Qualitative coding has a long tradition in social science research. It is a process of identifying themes, patterns, and insights in a corpus of text (Saldaña, 2011). Reflexive thematic analysis is one of the foundational methods in qualitative data analysis and is a method in which researchers identify, analyze, and report patterns (Braun and Clarke, 2006). These patterns, or themes, are findings in the data that answer the research question (Braun and Clarke, 2006). The benefits of thematic analysis are that it can be flexibly utilized in a broad range of approaches while providing detailed, complex findings. Following Braun and Clarke's tradition, thematic analysis has six phases: "1) familiarizing yourself with the data, 2) generating initial codes, 3) searching for themes, 4) reviewing themes, 5) defining and naming themes, and 6) producing the report" (Braun and Clarke, 2006, page 87). While the approach we describe here is not intended to replicate reflexive thematic analysis, it does describe aspects of thematic analysis vis-à-vis creating codebooks. Codebooks begin to be developed in phase two, refined in phases three through five, and used in phase six. A final codebook shows all the codes for an analyzed data set organized hierarchically to show the relationships between codes and descriptions and examples for each code (Gery and Bernard, 2000). This is just one example of qualitative coding; many more are outlined in popular qualitative coding manuals (Saldaña, 2021). Regardless of the tradition, there are non-trivial questions about what constitutes a code, differences between codes and themes, and which procedure is most advantageous for a given research question (Elliott, 2018).

Natural language processing and machine learning in thematic analysis. Researchers have long sought to identify themes in large corpora of texts using machine learning techniques, including topic modeling, clustering, and deep learning. For example, Blei et al. (2003) used the technique of Latent Dirichlet allocation (LDA) to identify topics in a corpus of text. A more modern version of LDA that uses transformer-based language models is BERTopic (Grootendorst, 2022). Before these deep neural network approaches, however, computer-assisted qualitative data analysis software was used for decades (Cope, 2014; Kelle and Bird, 1995). These tools (e.g., NVivo, ATLAS.ti, Dedoose) have been used to facilitate the coding process by providing a digital interface for researchers to organize and analyze their data. Other tools like quanteda (Benoit et al., 2018) or tidytext (Silge and Robinson, 2016) in R have been used to analyze text data in a more programmatic way but are not designed for thematic analysis per se and instead focus on text mining. Until the past few years, there were glimmers of potential for using machine learning techniques to facilitate thematic analysis, but computational resources and models were not yet sophisticated enough to approximate human-level results.

Qualitative researchers immediately began exploring the use of large language models (LLMs) for qualitative data analysis (QDA) following the release of ChatGPT in November 2022. ChatGPT is one of the most commonly used LLMs for QDA and has been used to study a variety of topics, including universal basic income, PhD students' transitions to being independent researchers, and

Table 1 Comparison of LLM-assisted thematic analysis workflows.		
De Paoli (2024)	Katz et al. (2023)	Gao et al. (2024)
Qual reference: Braun and Clark (2006) inductive thematic analysis 1. Familiarize with data: “researcher cleans the data, saves individual files in .txt format, creates interview chunks” 2a. Initial codes: “LLM identifies initial codes & quotes from interview chunks, it also provides their descriptions”. 2b. “LLM reduces duplicate codes, but keeps different quotes” 3. Searching themes: “LLM identifies themes and their descriptions by sorting and grouping initial codes and codes descriptions” 4. “Generate short labels of those summary groupings” 5. Defining themes: “LLM (re)names themes only using the codes, descriptions and quotes. Researcher confirms themes”	Qual reference: Braun and Clark (2017) thematic analysis 1. “Embed and cluster sentences” 2. “Generate 15-word summaries of those clusters” 3. “Cluster the summaries” 4. Reviewing themes: “LLM produces 3 new set of themes, using increased creativity (temp > =0.5). Research reviews for consistent themes for final selection”	Qual reference: Richards and Hempill (2018) grounded theory and thematic analysis 1. “Independent Open Coding, facilitated by on-demand code suggestions from LLMs, yielding initial codes” “2. Iterative Discussion, focusing on conflict mediation within the coding team, producing a list of agreed-upon code decisions” “3. Codebook Development, where code groups may be formed through LLM-generated suggestions, based on the list of decided codes”

engineering student career interests (Hamilton et al., 2023; Tai et al., 2024; Katz et al., 2023; Katz et al., 2024).

While some papers focus on the application of LLMs to specific use cases (i.e., datasets), others establish workflows for utilizing LLMs for QDA (De Paoli, 2024; Gao et al., 2024; Katz et al., 2023). We center our literature review on thematic analysis, this paper’s motivating method. De Paoli (2024) and Katz et al. (2023) focused on human-in-the-loop LLM-assisted thematic analysis with an individual human coder, while Gao et al. (2024) developed an LLM-assisted collaborative coding platform for multiple researchers. Table 1 shows a comparison of these three workflows and the approaches in which they are grounded.

Methods

This Methods section contains three subsections: 1) the data simulation process used to generate the synthetic datasets used in this study, 2) the central contribution of the paper: the GATOS workflow used to generate the codebooks for qualitative data analysis, and 3) the evaluation process. We present the methods in this order of data simulation before data analysis (GATOS workflow) because we believe the tradition of data collection (in the form of simulated data here) followed by data analysis might be most familiar to readers.

Data simulation. In this section, we describe how we simulated data for this validation study. The datasets were simulated to mimic the following three contexts that are encountered in social science research settings:

- 1. Teammate feedback
- 2. Organizational cultures of ethical behavior
- 3. Employee perspectives about returning to their offices after the pandemic

These contexts are pertinent to social science research to understand interpersonal teammate dynamics, organizations’ members’ perceptions of the organization’s culture of ethical behavior, and perspectives and attitudes toward workplace location policy shifts.

Our data simulation approach was inspired by common approaches to method development studies in quantitative research, wherein researchers generate synthetic data to test their methods (Morris et al., 2019; Burton et al., 2006). The philosophy is that if we control the data-generating process, then we know

what the method *should* identify as the correct answer. For quantitative methods, that might be the recovery of parameters used to generate the data. For qualitative methods, such as this paper, the analog is recovery of themes or sub-themes used to generate the text data. To be clear, we recognize there are multiple approaches to establishing validation evidence for methods, and we are using synthetic data as one of several approaches in an array of studies.

The objective for generating synthetic data was to generate data that were as realistic as possible, to data that one would encounter in live data collection. The realism here is an important element because even though these synthetic data were generated using LLMs, members of the research team reviewed the generated responses to check whether they sounded like realistic responses we have seen in prior studies. We achieved that objective of generating realistic responses through a multi-step process of first generating backstories and data generation criteria for the text model and then prompting generative text models to simulate data according to those generated criteria. Some criteria were generated by a generative text model while others were researcher-specified. Model-specified criteria included personas, contexts, themes, and sub-themes. Researcher-specified criteria included data type, data collection context, writing style, and writing length. These criteria were important for generating more realistic-sounding synthetic data. The process for data simulation is shown in Fig. 1.

The first step in the data simulation process was to generate personas. We used the open-source Llama3.1-70b generative text model to generate a set of personas each describing attributes of individuals (e.g., age, occupation, personality traits for each) that could be in each study. Then, for each dataset, we used Llama-3.1-70b to generate four imaginary contexts (i.e., situation or environment) based on the overall background that the data were supposed to come from. For example, a context might be “Government Contracting Agency”. Next, using Llama3.1-70b, we generated a list of eight themes that might appear in data for each study as well as eight sub-themes for each theme. We chose eight themes and eight sub-themes per theme to have enough variety without needing to generate too many simulated data points per sub-theme. We also manually specified writing lengths (e.g., short, medium, long) and writing styles (e.g., professional, casual, sentence fragments). Finally, we specified three models that could be used for the data generation process: Wizardlm2-8x22b (Xu et al., 2023), mistral-nemo-12b, and Llama-3.1-8b (Dubey et al., 2024). Each was chosen for its permissive license.

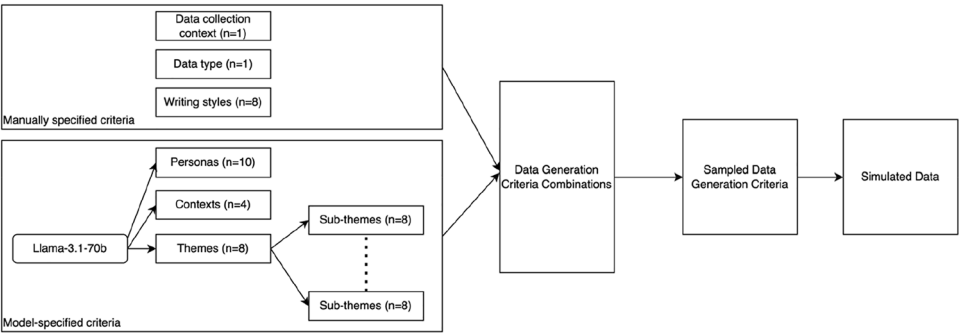


Fig. 1 Data generation process.

Table 2 Data generation criteria generated by text model.		
Data Generation Criterion	Criterion Description	Example
Personas	A set of attributes that describes an individual	Union Member
Persona Description	A description of a persona	A 50-year-old suburban resident working in manufacturing. He is a union member who values job security and fair labor practices, and believes that returning to the office should be subject to collective bargaining agreements. This persona may be cautious about returning to an office setting due to concerns about worker rights and protections.
Contexts	A description of a situation or environment	Government agency workplace
Context Description	A description of a context	A bureaucratic setting with strict protocols, hierarchical structures, and a focus on public service. Participants may express concerns about commuting time, office politics, and adapting to new health and safety regulations.
Themes	A list of themes that might appear in a study	Frustration with Lack of Clear Communication
Sub-themes	A list of sub-themes that might appear in a study under that theme; more specific than themes	Insufficient Information About Office Safety Protocols

Table 3 Manually specified data generation criteria.		
Data Generation Criterion	Criterion Description	Example
Data Type	The type of data being generated	Written responses
Data Collection Context	The context in which the data were collected	Employee perspectives on returning to work after the pandemic
Writing Style	The style of writing used in the data	Professional
Writing Length	The length of the written responses	Medium (4-5 sentences)
Model	The generative text model used to generate the data	Mistral-nemo-12b

Specifying this combination of models, personas, contexts, writing styles, writing lengths, themes, and sub-themes led to a combinatorially large number of hundreds of thousands of possible simulated data. Rather than generating millions of responses, we randomly sampled 18 possible data points for each sub-theme to generate the actual synthetic datasets used in this study. From a list of 64 sub-themes (8 themes with 8 sub-themes each), this theoretically would result in 1152 data points for each dataset. In practice, this number was smaller due to some overlap in sub-themes and issues with the data simulation process, wherein some models used for data generation did not follow instructions well; these data were discarded. We detected that some models did not follow instructions well in the following way: we instructed the LLM to produce its output in a JSON-structured format. When the format was not followed, then the parsing instructions failed. We then manually inspected those instances, noticed the failure to adhere to the instructions, and discarded those instances. We did not detect any patterns associated with when such failure cases occurred.

The types of data generation criteria used in this process that were generated by a language model are shown in Table 2, and the manually specified data generation criteria are shown in Table 3. The following subsections describe the data generation criteria and descriptive statistics for the three generated datasets.

Simulated Dataset 1: teammate feedback. The first synthetic dataset came from a hypothetical study of teammate feedback in settings where teamwork is essential. The specified context that the model was given was: “Teammate feedback surveys instructing students to respond to the prompt ‘Please provide constructive comments about your fellow teammates as well as yourself.’ The purpose of these comments is to give you the opportunity to explain how you rated your peers and if there was behavior or experience in particular that influenced you when doing your peer and self-evaluations”. After removing invalid responses, the simulated teammate feedback dataset consisted of 854 written responses. The average number of words was 194, and the median was 180. The

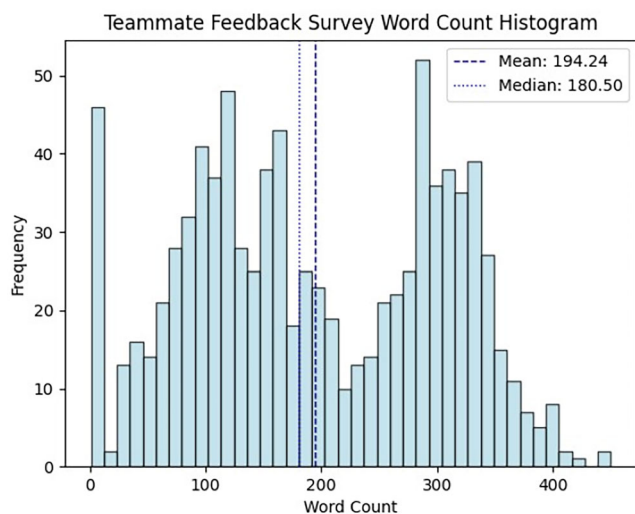


Fig. 2 Distribution of simulated response lengths for teammate feedback. Dashed line: mean (194.24), dotted line: median (180.50).

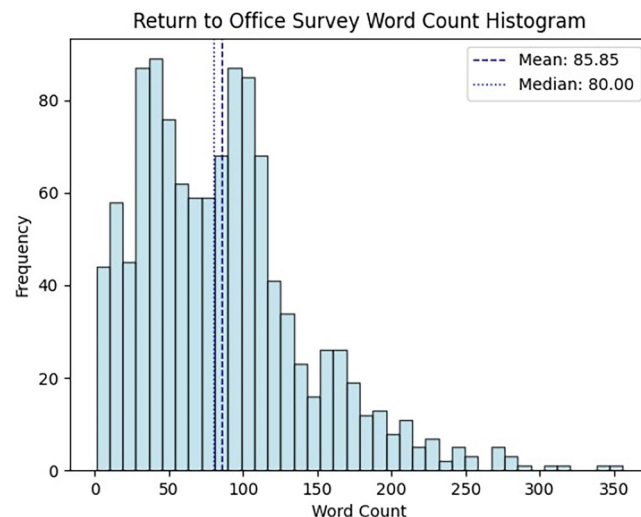


Fig. 4 Distribution of response lengths for student extracurricular activity participation. Dashed line: mean (85.85), dotted line: median (80.00).

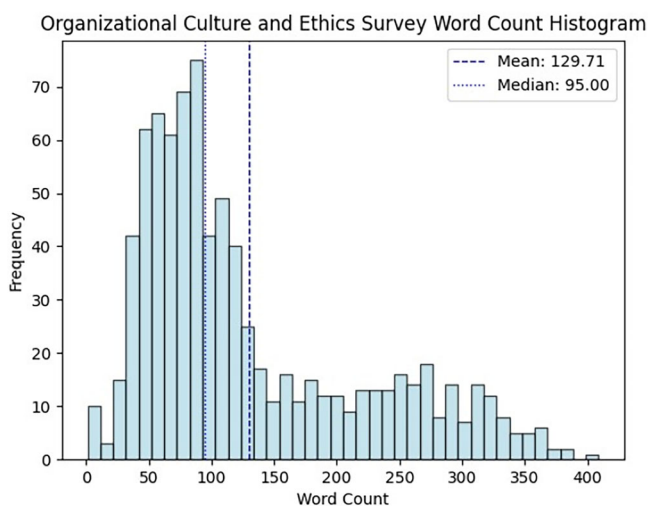


Fig. 3 Distribution of response lengths for faculty perspectives upon returning to work after the pandemic. Dashed line: mean (129.71), dotted line: median (95.00).

distribution of written response lengths, shown in Fig. 2, demonstrated a bimodal distribution. This distribution was likely due to the way the data were generated by forcing variety in response length through the writing style and writing length criteria.

An example of a simulated response focused on the theme “Conflict Resolution and Negotiation” and sub-theme “Effective Communication Strategies” is the following: “While I appreciated [Teammate’s Name]’s passion for environmental sustainability, their focus sometimes overshadowed the importance of social responsibility in our project. To resolve this, I found it helpful to ask open-ended questions and actively listen to their perspective. However, there were times when I could have been more assertive in expressing my concerns and proposing alternative solutions.” Another example response from the persona of a 25-year-old ambitious graduate student in this synthetic dataset about teammate feedback is given in Supplementary Material S.1.

Simulated Dataset 2: organizational cultures of ethical behavior. The second synthetic dataset is from a hypothetical study of organizational cultures of ethical behavior. The context for this imagined study was: “A study asking people working for large companies the following question: ‘What factors do you believe affect your organization’s culture around ethical behavior?’”

After removing invalid responses, there were 823 responses in this dataset. The average number of words in the responses was 129 and the median was 95. The distribution of the length of those written responses for this scenario, shown in Fig. 3, demonstrated a long-tailed distribution with a few responses being much longer than the rest. This was a deviation from the first dataset’s bimodal distribution.

An example of a 50-year-old union representative discussing the embedded theme “Communication Channels and Transparency” and sub-theme of “Accessibility of Ethics Resources” response was the following “In my experience, management loves throwing around fancy terms like ‘ethical culture,’ yet they fail miserably when it comes to making ethical guidelines accessible to us on the ground floor. The company’s code of conduct might as well be written in ancient hieroglyphics for all its relevance and readability! It’s buried deep within some obscure intranet site that no one knows about or cares to navigate through.”. An additional example response from the persona of a 32-year-old HR specialist in this synthetic dataset about organizational cultures of ethical behavior is given in Supplementary Material S.2.

Simulated dataset 3: employee perspectives about returning to their workplaces after the pandemic. The third synthetic dataset is from a hypothetical study of employee perspectives about returning to their offices after the COVID-19 pandemic. The specified context for this third study was: “A study or worker perspectives on returning to the office after the pandemic”.

The average number of words in the responses was 131 and the median was 97. The distribution of the length of the 1110 written responses for this scenario, shown in Fig. 4, demonstrated a more realistic long tail distribution like the second dataset, but with fewer responses being much longer than the rest. This distribution is more like our observations from prior work (Johnson et al. 2023; Katz et al., 2024; Paul et al., 2022), where we found that responses to open-ended survey questions often have a long-tailed distribution.

An example of a response focused on the theme “Nostalgia for Pre-Pandemic Work Environment” and sub-theme “Nostalgia for Office Culture and Traditions” that has a more varied writing style is the following: “I’m so ready to be back in the office... miss the energy of being around my colleagues all day... Friday team lunches at the new restaurant down the street... celebrating milestones and birthdays with cake and balloons... it’s those little things that made coming to work feel special... our team was like a family, always supporting each other... I hope we can get back to that soon... virtual happy hours just aren’t the same...”. An additional example simulated response from the persona of a working mother in this synthetic dataset about perspectives on returning to the office after the pandemic is in Supplementary Material S.3. We have provided these examples for the datasets to illustrate to the reader how the themes and sub-themes could manifest in the simulated data because the notion of an AI workflow evaluating AI-generated data may appear meaningless. Our suggestion is that these simulated data actually represent realistic responses because of how we embedded the themes, sub-themes, personas, contexts, and writing styles. This approach to using LLMs for generating simulated data is also an approach applied elsewhere in social science research (Argyle et al., 2023; Gao et al., 2024).

With these three synthetic datasets generated, we were able to test the GATOS workflow to see the extent to which the process could identify the themes and sub-themes used to generate the data.

GATOS workflow overview. The task we seek to solve with this method is to find recurring patterns in text data collected in realistic simulated research settings. Our solution was to combine several NLP tools and techniques with generative text models into a multi-step workflow. We use an open-weights generative text model Mistral-22b-2409 and modern NLP techniques (i.e., text embedding) to enable inductive qualitative data analysis at a large scale. By inductive, we mean a data-driven approach to generate a codebook rather than purely theory-driven. This is the kind of approach one may take when not knowing a priori what is discussed in the text. The researcher could also take a hybrid approach of providing their own codes as well based on their specific theoretical framework or research question in a more human-in-the-loop approach (Anakok et al., 2025). The generated codebook can then be used to label the original text units according to their themes. This application step is beyond the scope of the current paper and will be described in future work.

We break this inductive codebook generation process down into multiple parts. First, summarize the original text units of analysis (e.g., individual written responses to open-ended survey questions) in the context of the researchers’ specific research questions, which they provide as part of their instructions to the LLM. This produces a list of individual summary points analogous to observation memos that a research might make to summarize each observation they make as they read their data. Second, take all the atomic research-relevant summary points and use a text embedding model to generate high-dimensional numeric representations of each point. Third, reduce the dimensionality of those text embeddings to a lower-dimensional space to enable step four—clustering the embedded summary points. In theory, these clusters should contain semantically similar summary points. For example, summary points such as ‘frequent emails reminders’ and ‘constantly sent info to help stay on track’ could be clustered together since they are about the same idea (i.e., team leader communication about progress). At this point in the process, we have gone from an initial set of raw text units to a smaller set of clusters of semantically similar text

units; however, we still do not know what clusters are necessarily about—only that most of the clustered texts are somehow similar.

Up to now, many of these steps have been explored in prior research. The novelty of this workflow comes in step five. The goals of step five (codebook generation) are two-fold and contend with each other: (1) to generate a code for each cluster and (2) try to avoid generating too many redundant codes. This aims to mimic how a researcher might code data in a traditional qualitative data analysis setting. For example, the researcher might begin by reading their data and generating codes (i.e., short descriptive phrases accompanied by definitions, depending on their coding method) as they encounter new ideas. However, when they see data described by a code that already exists in their burgeoning codebook, they would not generate a new code and instead move on to the next piece of text. At each data point, therefore, the researcher performs multiple tasks: identify what is discussed in the data; think about whether it warrants a code; check whether a code already exists that would describe it; and, if not, create that new code and add it to the codebook.

In step five, we mimic that reflection process with the generative text model and retrieval augmented generation (RAG). We use a RAG approach as a way of managing the context that the LLM needs to attend to. Whereas we could provide the LLM the entire codebook to consider, in practice that might provide too much distracting information to the LLM. It is the analog of providing a researcher the k most relevant codes in the codebook to think about and make salient in their deliberation rather than making thinking about all the codes at once. Therefore, to operationalize this RAG approach here, we first take the cluster of text and find the k most semantically similar codes for each summary point in the cluster (usually, k is on the order of 2 to 4) using the embeddings for the summary points, embeddings for the existing codes, and calculating their cosine similarities. We then aggregate those nearest neighbor codes into a set, thereby removing redundancies, to generate a list of unique nearest neighbor codes for that cluster. For example, if there are six summary points in a cluster, we would find $6 * k$ nearest neighbor codes (from the growing codebook) for that cluster, though this list may have multiple copies of the same codes. If the six summary points are all about the team leader sending email reminders, we look for existing codes in the codebook that are most similar to this idea, but that may only result in two or three unique existing codes in the codebook. Those unique codes are then included in the prompt to the generative text model to decide whether to generate a new code for that cluster.

At a philosophical level, at this point in the process, it is still an open question whether those ‘most similar’ codes actually capture the idea(s) expressed in that cluster. This is analogous to a researcher having thought of similar codes in their codebook that might be applied to the data they are currently reading, but now needing to assess whether or not the code is appropriate to describe the data. To investigate this question, we instruct a generative text model to look at the cluster of (n) summary points, the existing codes in the codebook (as represented by the $\leq n * k$ nearest neighbor codes), and decide whether or not a new code is needed based on whether the existing codes in the codebook provide sufficient thematic coverage of the cluster of summary points. If the model decides a new code(s) is (are) needed, it generates a new code and definition for that code, which is added to the codebook. If the model decides a new code is not needed, the process simply moves on to the next cluster of data points. The full prompt for this code consideration step is in Supplementary material S.6.

By the end of step five, we have gone through each cluster of summary points and either identified a new code or decided that

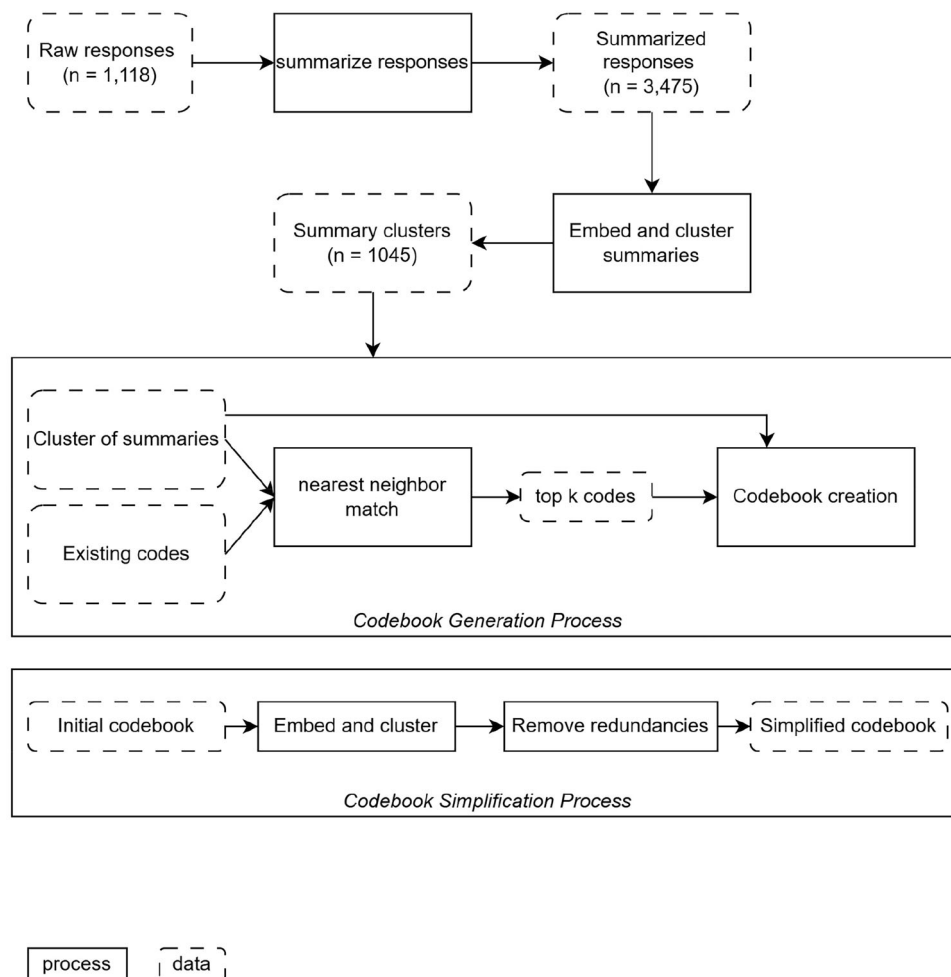


Fig. 5 Process for the GATOS workflow.

no new code was needed, thereby balancing the goals of identifying recurring patterns in the summary points while not generating too many redundant codes. In practice, there are still many near redundancies that exist, so the final step in is to cluster these newly generated codes and prompt a model to identify distinct themes. The process culminates with a list of themes and codes belonging to those themes. This entire workflow is shown in Fig. 5. We call this workflow the Generative AI-enabled Theme Organization and Structuring (GATOS) method because it is designed to mimic parts of thematic analysis while also being distinct from traditional thematic analysis.

For evaluation purposes in this study, the final step was to compare the themes generated through this process with the (8) themes and (8 * 8) sub-themes used to simulate the original data. The details of each step in the workflow and procedures to evaluate our workflow are described in more detail in Supplementary Material S.4.

Evaluation process. To evaluate the quality of the workflow's codebooks, we primarily used a manual approach because human judgment was required to assess the quality of matches. We did not use traditional machine learning metrics here (e.g., precision, recall) because the determination of whether there were good (defined momentarily) matches did not seem like it fit the model of a typical ML classification task. Instead, we framed the task as one where (for each of the three datasets) we had two sets of codes (our pre-defined/original sub-themes (i.e., codes), set A, and then the codes from the workflow, set B, and needed to

determine if there was a least one match from set B for each item in set A. In that framing, the task required human raters to determine what constituted a good match, where good meant the semantics of the code in set B matched the semantics of the code in set A. Conversely, a "bad" match meant that members of the research team did not identify a semantically similar code in set B for a code in set A. The three ratings for these matches were "good match", "partial match", and "no match". We performed this evaluation for the original sub-themes (compared with the generated codes) and original themes (compared with the generated themes) for each dataset. In addition to those manual ratings, we also calculated the semantic similarity between each sub-theme and the codes generated for that dataset so that we could plot semantic similarity scores between each sub-theme and its best match as additional weaker evidence of matches.

Results

We present the results of the simulation study across the three synthetic datasets. For each dataset, we compare the codebooks generated by the GATOS workflow with the codebooks used to generate the original data. Before reporting results from each case, we first give examples of input and output for each of the main prompts used in this process.

Example input and output for each generative step in GATOS workflow

Information extraction with a generative text model. An example of the input and output from the information extraction step is

Table 4 Example of information extraction from return-to-work synthetic dataset.

Input	Output
"I'm really excited about returning to the office, but I have to admit that I'm worried about how hybrid work arrangements will affect my career advancement. If we're not in the office regularly, I'll miss out on watercooler conversations with clients and colleagues, which can lead to new business opportunities. I've seen it happen before when people are working remotely, they tend to get stuck in their own little bubbles and don't have as many chances to connect with others. I'm hoping that our company will find a way to balance flexibility with face-to-face interaction, so we can all continue to grow and succeed together."	"My expert summary: 1. Excited about returning to the office 2. Concerned about career advancement in hybrid work arrangements 3. Fears missing out on networking opportunities with clients and colleagues 4. Believes remote work can lead to isolation and reduced connections 5. Hopes company balances flexibility with face-to-face interaction for collective growth"

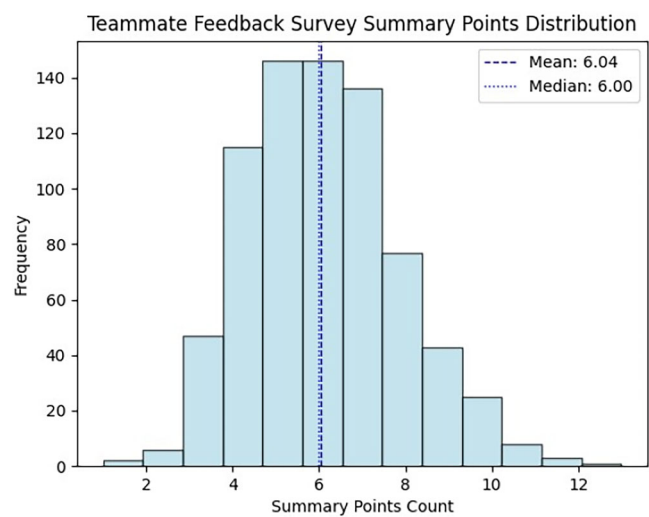


Fig. 6 Distribution of summary points per response for teammate feedback. Dashed line: mean (6.04), dotted line: median (6.00).

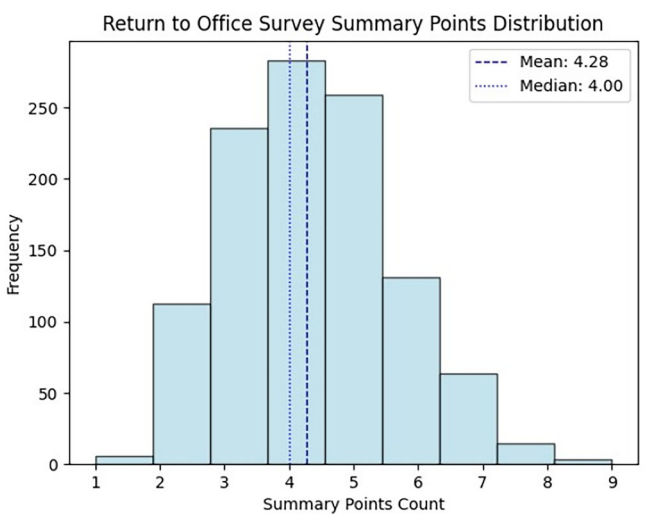


Fig. 8 Distribution of summary points per response for returning to work after the pandemic. Dashed line: mean (4.28), dotted line: median (4.00).

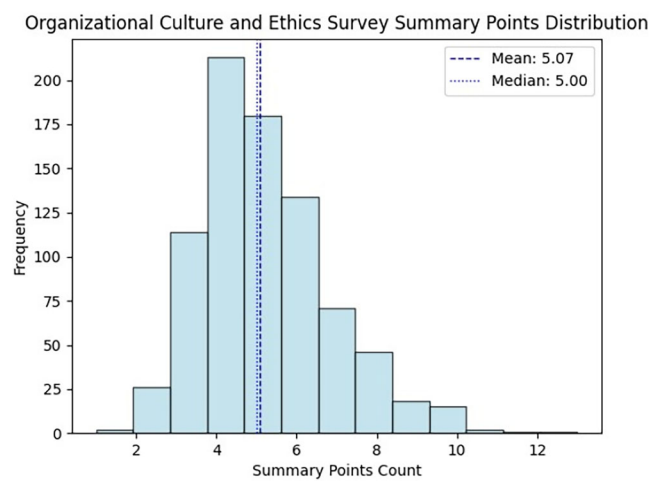


Fig. 7 Distribution of summary points per response for perspectives on organizational cultures of ethical behavior. Dashed line: mean (5.07), dotted line: median (7.00).

given in Table 4. We ran this step for each entry in the datasets to extract the key ideas from each response, which makes the data more manageable for the next steps. As shown here, the input is a medium-length simulated response from the return-to-office dataset, and the output is a five-point list of key ideas extracted

from that response. The specified theme and subtheme for the original synthesized data were “Expectations for Hybrid Work Arrangements” and “Fears about Negative Impact on Career Advancement”, respectively. Despite being given a single theme and sub-theme, the generative model included more than just those themes and sub-themes in the simulated response. This can be seen in the summary points extracted from the response, which include a mix of the specified theme and subtheme as well as additional information that was not part of the original specified criteria (but was in the simulated data as a byproduct of the simulation).

The distributions of the number of summary points extracted for the teammate feedback dataset is shown in Fig. 6, for the organizational culture of ethics synthetic dataset in Fig. 7, and for the return to workplace dataset in Fig. 8. The average number of summary points extracted from each response for the three datasets was 6.04, 5.07, and 4.28, respectively, showing how the original data (i.e., full response) can expand 4x–6x as it is broken into summary points for the next step in the process. Additionally, the distributions of these summary points more closely resembled normal distributions, which was a notable contrast compared to the long-tailed and bimodal distributions of the length of the original synthesized data.

Inductive codebook generation. After extracting information from each response to generate summary points, we clustered the summary points and prompted a generative text model to iteratively consider whether or not to create a new code to describe

Table 5 Example of theme identification from initial codebook.

Input Codes	Output Themes
1. Flexible work arrangements for wellbeing	1. Flexibility in Work Arrangements
2. Flexible work hours	2. Uncertainty in Work Schedules
3. In office work frequency and day uncertainty	3. Supportive Work Environment
4. Uncertainty about remote work frequency and future schedules	4. Clear Guidelines and Boundaries
5. Flexible work arrangements for caregivers	5. Control Over Work Hours
6. Individualized flexible work arrangements	
7. Supportive, inclusive work environment	
8. Resistance to traditional office hours	
9. Staggered work hours/shifts	
10. Clear workday end time and personal time protection guidelines	
11. Break and lunch dynamics uncertainty	
12. Control over work hours	

that cluster. In the codebook creation prompt, the codes in the existing codebook XML tag in the prompt were the nearest codes to the new cluster of summary points. At this point in the GATOS workflow, there are two possible outcomes from the generative text model: (1) a new code(s) is generated and added to the codebook or (2) no new code is generated. The following three gray boxes present the input and output for the inductive codebook generation step where a code is created when analyzing the return to work after the pandemic dataset. Following this example, where a code is suggested, we also provide an example where the recommendation is to not create a new code. The input summary points for the cluster were: 1) requires seamless video conferencing for communication; 2) desires user-friendly video conferencing tools; 3) up-to-date hardware and software solutions required for smooth transitions; 4) requires robust infrastructure for seamless video conferencing and secure VPN access at home; 5) hopes for reliable video conferencing tools (Zoom, Google Meet); 6) wants seamless communication platforms (Slack, Microsoft Teams); 7) specific technological needs: high-quality video conferencing, secure VPNs, cloud-based software applications; 8) request for reliable video conferencing tools at home; 9) reliable video conferencing and high-speed internet are necessary; 10) unfamiliar with potential new collaboration tools. We see that many of the summary points in this cluster are about the need for reliable video conferencing tools and secure VPN access for remote work. The nearest neighbor codes for this cluster were: 1) ability to change, 2) social interaction needs, 3) adaptability to sudden changes, 4) productivity at home, 5) anticipation of social interaction, 6) training and support needs, 7) information clarity frustration, 8) technological challenges, 9) flexible scheduling and remote work options, 10) navigating unpredictable changes.

These nearest neighbor codes discuss changes, productivity at home, social interaction, and technological challenges. From these summary points and nearest neighbor codes, the model generated the reasoning and concluding code recommendation given in Supplementary Material S.8.

As this example shows, the model proceeds through each of the instructed steps, starting with a consideration of the existing codes in the codebook. The model then reads the cluster's summary points and identifies the main theme as the need for reliable video conferencing tools and secure VPN access. The model then considers the existing codes in the codebook and finds that the code "technological challenges" partially captures the main theme but does not fully encompass the specific needs mentioned in the response. After more reasoning steps, the model decides that a new code is needed to capture the specific technological needs expressed in the written response. The code suggested was "reliable video conferencing tools", which matches some of the main ideas in the summary points yet was not already

covered in any existing codes in the codebook. Thus, the generative text model displayed a step-by-step reasoning sequence that culminated in a new code being added to the codebook. In contrast with this example, the next example, also from the return to work after the pandemic dataset, shows an instance where the model decided not to create a new code. *Input Summary Points:* 1) open to hybrid work arrangements for balancing work and family commitments, 2) hope company takes concerns seriously and works together on beneficial hybrid arrangements, 3) need for hybrid work arrangements that accommodate various needs (e.g., caring for elderly parents or young kids), 4) prefers flexible, hybrid work arrangement for balancing career and family responsibilities, 5) hopes for hybrid work model to balance family and job responsibilities.

This cluster of summary points is about the need for hybrid work arrangements that accommodate various personal needs, such as caring for elderly parents or young kids. The nearest neighbor codes for this cluster were: 1) work-life balance, 2) team integration anxiety, 3) flexible work hours, 4) hybrid work flexibility, 5) single parent work-life balance, 6) job security fears, 7) childcare arrangements. The nearest neighbor codes for this cluster of summary points centered around work-life balance, hybrid work flexibility, and family responsibilities.

In this example (output in Supplementary Material S.9), the model decided that no new code was needed because the existing codes in the codebook were sufficient. This response seems plausible because the code "hybrid work flexibility" sufficiently covers the central idea in the cluster of summary points, namely the need for flexible, hybrid work arrangements that accommodate various personal needs. This example illustrates the central point of this entire process: balancing the need for new codes to capture the nuances of the data with the need to avoid generating redundant codes. Moreover, this response illustrates how the model can make a reasoned decision like what a human would do when creating a codebook inductively.

Theme creation. An example of the input cluster of codes and output suggested themes from the theme creation step is given in Table 5. The input codes contained some redundancies related to flexible work arrangements and work hours, so the generative text model identified those as part of the theme "Flexibility in Work Arrangements". The model also identified themes related to uncertainty in work schedules and breaks, so those were combined into the theme "Uncertainty in Work Schedules". This example also illustrates how individual codes could also persist in the theme identification process, as in the case of the code "Control Over Work Hours". The example also illustrates how there is a thin line between a code and a theme, as the code "Control Over Work Hours" could be seen as a theme on its own

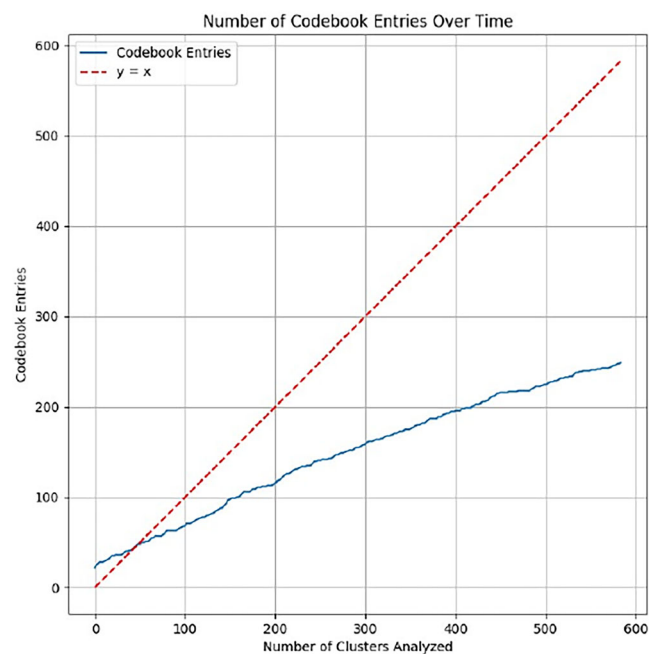


Fig. 9 Codes generated vs clusters analyzed for teammate feedback. Solid blue line: number of codebook entries, dashed red line: $y = x$. Solid blue line: number of codebook entries, dashed red line: $y = x$.

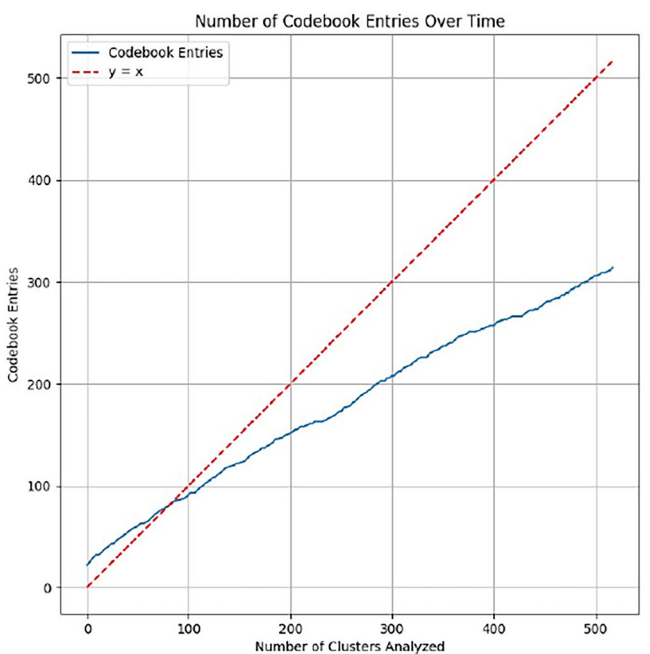


Fig. 11 Codes generated vs clusters analyzed for returning to work after the pandemic. Solid blue line: number of codebook entries, dashed red line: $y = x$.

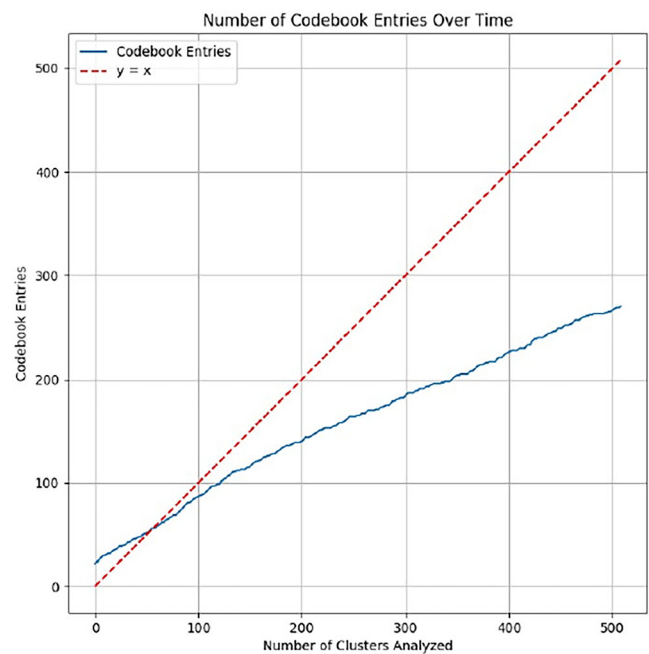


Fig. 10 Codes generated vs clusters analyzed for organizational culture of ethics. Solid blue line: number of codebook entries, dashed red line: $y = x$.

or could be subsumed into the theme “Flexibility in Work Arrangements”.

Codebook creation rates. One important objective of the GATOS workflow is to generate codes that capture the ideas in the data while avoiding redundancy in the codes. To track this, we noted the rate of code creation over time as each cluster was scrutinized by the generative text model. In the worst-case scenario, the model would create one (or more) new codes with each cluster it encountered. This would be unideal because there would

Table 6 Summary of GATOS workflow.				
Dataset	Data Points ^a	Summary Points	Codes ^b	Themes
Teammate Feedback	854	4563	249	89
Organizational Culture of Ethics	823	4176	246	75
Return to Workplace	1110	4748	314	110

^aSome observations dropped due to parsing errors or duplication.
^bNumber of codes includes original 20 codes generated by model to initiate process.

almost certainly be thematically redundant clusters, leading to redundant codes being created and defeating the purpose of the codebook checking step. Instead, if the model actually were checking the codes in the codebook to decide whether to create a new code or not, we would expect to see the model generate fewer new codes over time as it encounters more clusters and the codebook becomes saturated. This would suggest that the model is finding existing codes in the codebook that can describe the ideas in the new clusters. The rate of code creation for the teammate feedback dataset is shown in Fig. 9, for the organizational culture of ethics dataset in Fig. 10, and for the return-to-workplace dataset in Fig. 11.

As the figures show, the GATOS workflow met the minimal success criterion of not always generating a new code for each cluster. This can be seen by the deviation of the blue line from the red 45-degree line in the figures. The blue line represents the cumulative number of new codes generated by the model, while the 45-degree line represents the trajectory one would expect to see if there were a new code created for each cluster. The blue line bending below the red line is a quick way to see that sometimes the model does not create a new code for a cluster. The figures also show how many overall codes are initially created for the three datasets. Table 6 shows a summary of the starting number of simulated data points, summary points, codes, and themes for each dataset.

Table 7 Best examples of comparison of codebooks for teammate feedback synthetic dataset.

Original synthetic sub-theme	GATOS workflow method closest match
effective use of feedback mechanisms	feedback mechanism effectiveness
empathy and understanding in communication	empathetic communication
leadership and initiative taking	leadership and initiative
effective communication styles	effective communication
embracing feedback as a learning opportunity	openness to feedback

Table 8 Worst comparisons of codebooks for teammate feedback synthetic dataset.

Original Synthetic Sub-Theme	GATOS Workflow Method Closest Match(es)
Setting realistic goals for self-improvement	personal growth; learning and growth; self-reflection
Clear and concise messaging	effective communication; communication effectiveness
Motivates teammates to work towards common goals	effective team dynamics; mutual support and collaboration
Avoiding blame	personal bias management; negative behaviors and their impact

For the teammate feedback dataset, from an initial 854 simulated responses, the workflow generated 249 codes and 89 themes. For the organizational culture of ethics dataset, from an initial 823 simulated responses to the question of factors influencing their organization’s culture of ethical behavior, the workflow generated 246 codes and 75 themes. Finally, for the simulated dataset for expectations for returning to the workplace after the pandemic, the workflow generated 314 codes and 110 themes from an initial 1110 simulated responses. In each case, there is approximately a 10:1 ratio of simulated responses to themes generated by the GATOS workflow. It is an open question whether this is a coincidence or a generalizable pattern. At this point in the process, it is also an open question whether the themes generated by the GATOS workflow matched the sub-themes used to generate the original synthetic data. We address this question in the next section.

Evaluation: comparing original themes and sub-themes with GATOS-generated themes. In this section, we present the answer to the central question of this paper: *how well do the codebooks generated through the GATOS workflow capture the themes and sub-themes used to generate the original synthetic data?* The idea for this comparison is basic: compare the generated themes from the workflow with the themes and sub-themes that were intentionally used to generate the simulated data. We present the answer to this question for each of the three datasets in their respective subsections. In each subsection, we present examples where the workflow nearly matches the original sub-themes one-to-one and examples where the workflow only partially captures the original sub-theme. It is important to note that we are only presenting the comparisons for the GATOS workflow-generated themes and the original sub-themes. We focus on the comparison between generated themes and original sub-themes because in each of the three datasets, each of the original eight themes was recovered with a good match with at least one generated theme. For example, the original theme “leadership tone and accountability” matched the generated theme “leadership tone and impact”. This pattern of good matches was consistent across all original themes for the three datasets, so we do not explore those matches further in the results. Additionally, we do not present the comparisons for the individual codes generated by the GATOS workflow because the codes are not the final output of the workflow; however, we did find in our testing that those codes always matched an original sub-theme perfectly. One might expect that result given the number of codes generated by the

workflow (in the range of 246 to 314, as shown in Table 6) and the number of sub-themes used to generate the original synthetic data.

Synthetic Dataset 1: teammate feedback. There were 60 sub-themes (after removing a few redundant sub-themes) used to generate the original synthetic data for the teammate feedback dataset. The codebook generated by the GATOS workflow contained 249 codes and 89 themes. An example comparison of the codebook generated by the GATOS workflow with the original sub-themes is shown in Table 7. In this table, we selected the best examples (as determined by proximity of the GATOS theme with the original sub-theme) of how the GATOS workflow themes aligned with the sub-themes in the original synthetic data. There were many examples where the GATOS workflow-generated themes closely matched the original sub-themes. For example, the original sub-theme “effective use of feedback mechanisms” most closely matched the GATOS workflow theme “feedback mechanism effectiveness”. Another near-perfect match was for the sub-theme “embracing feedback as a learning opportunity”, which matched the GATOS workflow theme “openness to feedback”. The latter example demonstrates how the GATOS workflow can capture the semantic meaning of a sub-theme without using the exact same words (i.e., openness and embracing).

We also present the worst matches (i.e., where there were no themes from the workflow that matched an original sub-theme from the data generation process) or unmatched themes for the teammate feedback synthetic dataset in Table 8, showing the instances where there was no clear theme that perfectly aligned with a subtheme used in the original data generation step. For example, the simulated sub-theme “setting realistic goals for self-improvement” was only matched by “personal growth” and “self-reflection” in the GATOS-generated themes. Those themes are missing an aspect of goal setting that was present in the original sub-theme. The worst match of these poor matches was for the sub-theme “avoiding blame”, which was matched with the themes “personal bias management” and “negative behaviors and their impact”. While these themes might indirectly relate to avoiding blame, they miss the essence of blame avoidance. It should be noted that the original codes generated by the GATOS workflow (before they were combined into themes) did indeed have a closer match of “dismissive behavior”, though this, too, misses the aspect of blame. In total, 54 of the original sub-themes had at least one good match with the GATOS workflow-generated themes, and six of the original sub-themes had partial matches. There were no original sub-themes without at least a partial

match. The distribution of semantic similarity scores for the best match of the generated codes for each original sub-theme, grouped by “no match”, “partial match”, and “good match” is shown in Fig. 12. This figure shows that on average there were lower max similarity scores for partial matches compared to good matches, as one would expect a priori.

Synthetic Dataset 2: organizational culture of ethics synthetic dataset. With the synthetic dataset of organizational cultures of ethical behavior, there were 64 subthemes used to generate the original synthetic data. The codebook generated by the GATOS workflow contained 246 codes and 75 themes. An example comparison of the codebook generated by the GATOS workflow with the original sub-themes is shown in Table 9. As with the first dataset, for this initial table we selected the best examples of GATOS workflow themes most clearly aligned with the original synthetic dataset’s sub-themes. For example, the original sub-theme “visible consequences for leadership misconduct” most closely matched GATOS workflow theme “leadership misconduct and consequences”. Another near-perfect match was for the sub-theme “regulatory environment constraints”, which matched GATOS workflow theme “regulatory environment dynamics”.

Table 10 shows the instances where there were no ideal matches between the GATOS workflow themes and the original sub-themes. Examples of the worst fit between the GATOS workflow-generated themes and original sub-themes included “financial rewards for whistleblowing”, and “media scrutiny and public opinion”. The nearest matches for the original whistleblower sub-theme alluded to financial rewards whereas the workflow-generated theme was more general about whistleblowing and retaliation and omitted reference to rewards. Likewise, the media scrutiny and public opinion sub-theme was matched with the themes of 1) public perception and image and 2) media and public relations, which were close but not individually exact

matches; however, together the two workflow-generated themes could capture the central idea of the original sub-theme. In total, of the 64 sub-themes, three had no match, nine had a partial match, and the remaining 52 had good matches. Most often, the partial matches were missing some aspect of a sub-theme as in the example of the original sub-theme “goals overriding ethical considerations” having a closest code of “prioritization of non-ethical factors” where it was not clear that the goals were what was being prioritized. The distribution of semantic similarity scores for the best match of the generated codes for each original sub-theme, grouped by “no match”, “partial match”, and “good match” is shown in Fig. 13. As with Fig. 12, this figure shows that on average there were lower max similarity scores for the “no match” category compared with the “partial match” category, which in turn was lower on average than the good matches category. Again, this is consistent with a priori expectations.

Synthetic Dataset 3: perspectives about returning to workplace after the pandemic dataset. Finally, for the synthetic dataset on returning to the workplace after the pandemic, there were 63 sub-themes used to generate the original synthetic data. The GATOS workflow-generated codebook generated contained 314 codes and 110 themes. Table 11 shows an example comparison of the codebook generated by the GATOS workflow with the original sub-themes. As with the other two datasets, there were several examples of near-perfect matches between the original sub-themes and the GATOS workflow-generated themes. For example, the original sub-theme “resistance to traditional office hours” most closely matched the GATOS workflow theme “resistance to traditional office hours”. Another near-perfect match was for the sub-theme “concerns about micromanaging and surveillance”, which matched the GATOS workflow theme “micromanagement and surveillance concerns”. In those instances, even the syntax of the sub-theme was captured in the GATOS workflow-generated theme along with the semantics.

Table 12 presents the instances where there were no ideal matches between the GATOS workflow themes and the original sub-themes. Unlike the first two synthetic datasets, however, there were no instances where the GATOS workflow themes did not match the original sub-themes. The worst match was for the subtheme “confusion over new policies and procedures”, which was matched with the GATOS workflow theme “uncertainty about specific post-pandemic office aspects”. Those are close matches, with the difference possibly being the lack of confusion being mentioned in the GATOS workflow theme. The second sub-optimal match was for the sub-theme “worry about being judged or evaluated”, which was matched with the GATOS workflow themes “social expectation pressure anxiety”. The aspect possibly missing in the workflow-generated theme is an element of judgment. In total, there were 60 good matches and three partial matches for the sub-themes with the generated codes. As with the prior two datasets, we show the distribution of semantic similarity scores for the best match of the generated codes for each original sub-theme, grouped by “no match”, “partial match”, and “good match” in Fig. 14.

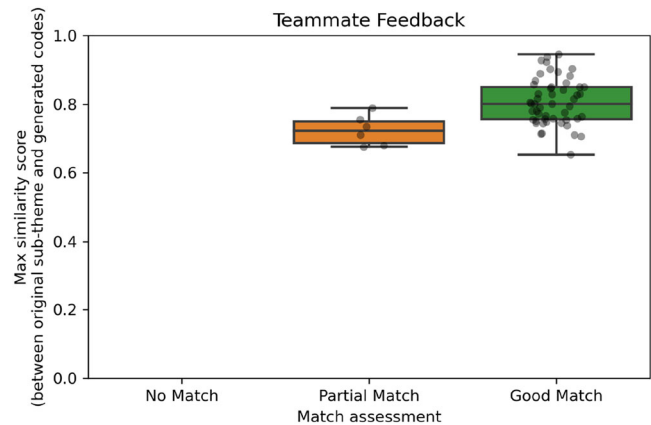


Fig. 12 Semantic similarity scores for top generated code match to each original sub-theme for the three evaluation categories for teammate feedback dataset.

Table 9 Best comparisons of codebooks for organizational culture of ethics synthetic dataset.	
Original Synthetic Sub-Theme	GATOS Workflow Method Closest Match
visible consequences for leadership misconduct	leadership misconduct and consequences
addressing microaggressions and harassment	harassment and microaggressions
representation in leadership positions	diversity and representation in leadership
regulatory environment constraints	regulatory environment dynamics
prioritization of ethics in performance evaluations	integration of ethics into performance evaluations

Table 10 Worst comparisons of codebooks for organizational culture of ethics synthetic dataset.

Original Synthetic Sub-Theme	GATOS Workflow Method Closest Match
Financial rewards for whistleblowing Media scrutiny and public opinion	Whistleblowing and retaliation; reporting unethical behavior public perception and image; media and public relations

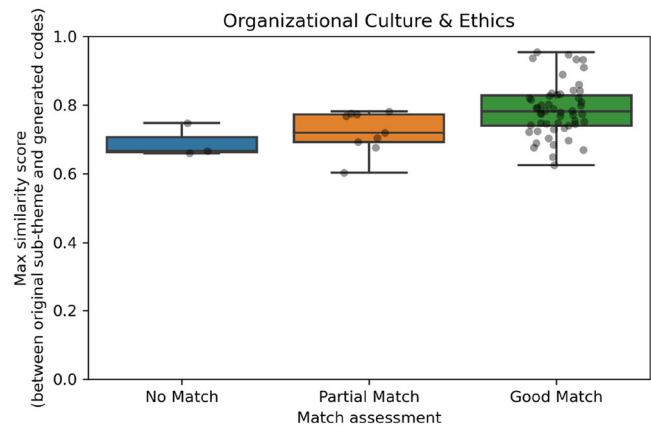


Fig. 13 Semantic similarity scores for top generated code match to each original sub-theme for the three evaluation categories for organizational culture of ethics dataset.

Discussion

Our investigations of the GATOS workflow in this simulation study show that the method can generate themes that closely match the sub-themes used to generate the original synthetic data. This process can be used to facilitate the creation of qualitative codebooks by augmenting human researchers’ abilities to analyze large corpora of qualitative data, as the results suggest. Researchers may want to do this when interested in making inferential claims to identify potential differences in experiences across groups. One simple example is a recent study that examined differences among underrepresented students in team settings and the kinds of feedback they receive (Dickerson et al., 2024). The authors in that study had access to a dataset of hundreds of thousands of student teammate feedback comments. In such settings, the workflow we described here could be used to create codebooks in a scalable way.

In our study, we borrowed from the practice in quantitative research methods development in which researchers generate synthetic datasets to test the performance of a new method by observing whether the new method captures the underlying data generation process (Morris et al., 2019; Burton et al., 2006). In our case, we generated synthetic qualitative data with known themes and sub-themes to test the GATOS workflow. In all three datasets, the GATOS workflow-generated themes were able to capture the essence of the original sub-themes. Fewer than 5% of the original sub-themes did not have a clear match, and under 2% had no close match. If these results hold in other contexts and datasets from human participants (instead of synthetic datasets), the GATOS workflow could be a valuable tool for researchers to use in qualitative data analysis across myriad social science and humanities research contexts.

The plots of code generation rates over time (Figs. 9, 10, and 11) show that the GATOS workflow was able to generate fewer new codes as it encountered more clusters. For us, this was only a recently observed phenomena with newer generative text models because prior models would not consistently generate internally coherent reasoning traces to decide whether to create a new code. Anecdotally, in our experience with prior versions of generative

text models trained on fewer data, models displayed faulty reasoning 10–20% of the time, an error rate unacceptable for research applications. However, newer generations of models (e.g., Mistral-small-22b) have improved to the point where they can consistently generate plausible and consistent reasoning. This level of reliability is crucial for any method that is intended to be used in research (Clonts, 1992; Golafshani, 2003).

Comparison with other studies using generative text models for qualitative data analysis. The idea of using generative text models to facilitate qualitative data analysis is not new. Yet, the GATOS workflow is unique in several ways. First, the GATOS workflow uses open-source (or at least open-weights, see Liesenfeld and Dingemanse, 2024 for a discussion on the distinctions) generative text models, which makes the method more accessible to researchers. Many prior studies have used proprietary generative text models (e.g., ChatGPT), which can be expensive and difficult to access. Using those proprietary models also raises important ethical concerns regarding data privacy and security (Davison et al., 2024). Many of these concerns arise from sending data to third-party model providers where the research team no longer has control over the data. Instead, using local, permissively licensed (e.g., Apache-2.0 license) LLMs can help to mitigate those concerns by avoiding such data transmission. Using open-weights models also enables better control over which model is used. To wit, when using proprietary models, researchers are often limited to the models that are available from the provider, which can be updated or changed without a researcher realizing the change. This is detrimental to the reproducibility of the research because different models yield different results, as evidenced by constant leaderboard changes and performance metric saturation (Ott et al., 2022). When an upgrade happens to a model used in a research study, this might raise issues similar to inter-rater reliability with human research teams (Belotto, 2018). Although these language models are probabilistic and might have seemingly inherent limitations in their ability to provide consistency, we addressed this issue by setting the model temperature to 0. This setting forces a more deterministic output for a model when given the same input.

Second, our approach develops an entire workflow for qualitative data analysis, from initial summarization to theme creation. Many prior studies have only used generative text models for one part of the qualitative data analysis process, such as coding or summarization. For example, Prescott et al. (2024) simply sent their data to ChatGPT and Bard. In a different study, Perkins and Roe (2024) used a human-in-the-loop approach to work with ChatGPT through multiple rounds of prompting to assist their inductive qualitative analysis. Qiao et al. (2024) took yet a different approach and fine-tuned ChatGPT before then sending their data to their fine-tuned ChatGPT to perform various steps of thematic analysis. The GATOS workflow is different from these approaches in several regards. First, the workflow uses a process more common in qualitative codebook generation with several steps that align with Braun and Clarke’s (2006) steps for thematic analysis. Second, the workflow uses a generative text model at multiple steps (summarization, codebook generation, and theme creation) along with an embedding model to add some structure to otherwise unstructured data. In contrast

Table 11 Best comparisons of codebooks for returning to workplace after the pandemic synthetic dataset.	
Original Synthetic Sub-Theme	GATOS Workflow Method
resistance to traditional office hours concerns about micromanaging and surveillance uncertainty about emergency response plans and protocols more comfortable and personalized workspace nostalgia for office culture and traditions	resistance to traditional office hours micromanagement and surveillance concerns emergency response plan uncertainty comfortable workspace customization nostalgia for pre-pandemic office culture and traditions

Table 12 Worst comparisons of codebooks for returning to workplace after the pandemic synthetic dataset.	
Original Synthetic Sub-Theme	GATOS Workflow Method Closest Match
confusion over new policies and procedures worry about being judged or evaluated	uncertainty about specific post-pandemic office aspects social expectation pressure anxiety

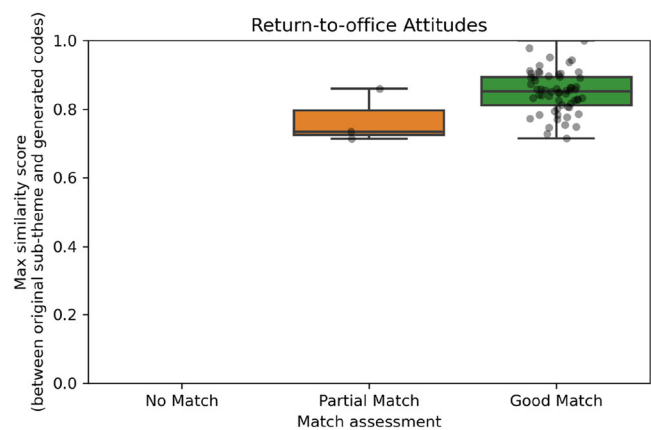


Fig. 14 Semantic similarity scores for top generated code match to each original sub-theme for the three evaluation categories for return-to-office attitudes dataset.

with other approaches that use embeddings to facilitate identifying topics in large corpora of text, e.g., BERTopic (Grootendorst, 2022), the embeddings are simply one part of a much larger process to leverage LLMs in the structured workflow. This structuring is important for the workflow to be able to generate themes that are coherent and meaningful and capture recurring patterns in the data. The method also aims to generate non-redundant codes that are collectively exhaustive yet mutually exclusive, which topic models or other methods typically do not achieve. In other studies that have used ChatGPT or other LLMs to generate code, there are no measures in place to try to make sure the LLM did not miss any noteworthy themes.

Third, referring to the point in the preceding paragraph, the GATOS workflow uses open-weight generative text models rather than ChatGPT (or other proprietary models), which makes the method more accessible to researchers and alleviates some concerns about data security and privacy. While some researchers advocate for using ChatGPT in social science research (Salah et al., 2023), we caution against that approach because of the aforementioned concerns about data privacy and model transparency.

Another contribution of our work, compared to other recent studies using ChatGPT for qualitative data analysis, is our use of synthetic datasets to demonstrate the utility of the qualitative text analysis method. Generating and testing a qualitative method on simulated data is an approach that allows researchers to test the performance of the method in a controlled setting. Simulation studies are common in quantitative research methods development (Morris et al., 2019), but they are much less common in

qualitative research methods development. This might, in part, be due to ontological and epistemological assumptions associated with qualitative paradigms, which often emphasize the importance of context and the situatedness of knowledge (Guba, Lincoln, et al., 1994; Lincoln et al., 2011). As others have noted, ontology and epistemology precede methodology (Grix, 2002). We do not mean to suggest the GATOS workflow will be most consistent with some humanities and social science researchers’ worldviews; yet, for those interested in computer-assisted approaches or researchers grappling with large volumes of qualitative data, the GATOS workflow could be a valuable tool.

Along with mimicking the simulation study paradigm from quantitative research traditions, using synthetic data also offers another advantage: working across a broader range of contexts and data types for testing the method. Many of the extant studies using generative text models for qualitative data analysis have worked on much smaller scales with smaller amounts of data and rarely in more than one context. Prescott et al. (2024) used 40 SMS text messages, Qiao et al. (2024) used nine interviews, and Perkins and Roe (2024) used 28 policy documents—all relatively small datasets. The GATOS workflow study here, on the other hand, was able to work with datasets of 854, 823, and 1110 observations from three distinct contexts. This is a much different scale than many other studies using generative text models for qualitative data analysis. In our view, the ability to work at this scale is one of the main reasons to use computer-assisted analysis techniques both for validation-type studies that leverage synthetic data but also more generally when using the method itself. When conducting social science research, working at that scale can be resource-intensive for a team of human researchers. The GATOS workflow can help alleviate some of those resource constraints and enable researchers to work with larger datasets more efficiently. In turn, working at that scale can allow researchers to answer more systematic questions about organizational phenomena, thereby trying to find a middle ground between the richness of qualitative traditions and the statistical power of quantitative traditions.

There are several natural extensions to the work we presented here, introducing the GATOS workflow. Some of the most immediate future research directions include testing other datasets that have already been analyzed in a more traditional qualitative coding process; testing other language models, embedding models, and cluster analysis methods for applying the codebook; testing the scalability of the method to larger datasets; and testing the method on other data with more complex structures (e.g., long interview transcripts). The first extension—trying to use the GATOS workflow to replicate human analysis on live datasets—would be a good test of the method’s generalizability and the extent to which it can mimic what human researchers had previously done. Indeed,

this study is currently underway by the authors on three datasets collected from human participants in educational research settings.

The second extension—testing other language models, embedding models, and cluster analysis methods—would be a good test of the method's flexibility and robustness, akin to sensitivity analyses. In typical quantitative research sensitivity analyses, researchers test the robustness of their results to different assumptions or methods (Saltelli et al., 2000; Iooss and Lemaître, 2015). When complex systems and models are being used for analysis, there are often myriad assumptions and parameters that could be affecting the output. Systematically testing the method with different language models, embedding models, and cluster analysis methods would help to ensure that the method is robust to those choices. Such analysis could also function as an optimization search for the best combination of models and methods to use in the workflow.

Finally, testing the model on larger quantities of data and more complex data structures would help identify the boundaries of where the GATOS workflow does and does not work well. For example, the simulated data tested for this study were relatively short responses to a single question. Such data tend to represent a small number of topics. In contrast, semi-structured interviews can explore a much larger topic space, depending on the phenomena investigated and questions asked. In our initial testing, we have explored segmenting interviews into smaller units of analysis, but we have not yet tested the GATOS workflow on full interview transcripts.

Limitations. The present study contains several limitations. First, the work we presented here utilized synthetic data for validation. There is a tradeoff in that approach. On the one hand, this allows for control over the “ground truth” that the workflow should recover. We presented this as a significant contribution earlier in this Discussion section. On the other hand, the synthetic data may not fully capture the complexity and nuances of human-generated qualitative data. Additionally, there could be something about the way the data were generated that keeps them in distribution for subsequent language models to summarize, embed, and analyze, which is unlike live data. To mitigate part of this issue, we used different language models for the data synthesis and codebook generation tasks. Specifying personas, contexts, and writing styles in the data generation criteria was also intended to help this issue. We also read a subset of each synthetic dataset to ensure that the data were coherent and realistic. Second, generative text models have been noted to have some biases when completing selective tasks (Liang et al., 2021). This is especially noted in generative text-to-image models (Bird et al., 2023). While we did not observe any biases in the generative text models we used, it is possible that biases emerge in other contexts.

Regarding the workflow itself, there are other notable limitations. First, the workflow still generates nearly redundant code sometimes. The move from codes to themes helps to reduce those redundancies, but they occasionally persist nonetheless. The extent to which this is a concern depends on the use case. Second, there are open questions about the appropriate level of abstraction for the code generation. This is an area where we believe a human-in-the-loop approach can help the workflow and refine the final codebook. Third, there remain open questions about the scalability of the method. The steps in the GATOS workflow can be computationally intensive when working with tens of thousands of observations. For example, imagine a dataset of 10,000 comments from a large-scale study of corporate culture. The GATOS workflow would need to first summarize each of those comments. If we assume that each comment leads to five summary points, we now have 50,000 summary points to cluster. Assuming each cluster has 10 summary points, that still leads to 5000 clusters for the

model to analyze when determining whether to create a new code. This is a computationally intensive process, and it is not clear how well the method would scale to even larger datasets. One way to address this issue would be to use iterative rounds of clustering the summary points to reduce instances where there are clear repetitions of summary points because people said the same thing. Finally, this process does not incorporate many opportunities for iterative development of the codebook as it is a primarily linear process. We believe incorporating iterative self-improvement steps would be an important area for future development.

Conclusion

In this paper, we have presented a method to use open-weights NLP tools and generative text models to mimic thematic analysis of qualitative data. We call this method the GATOS workflow. To test the ability of the GATOS workflow to accurately capture themes and sub-themes in qualitative data, we generated three synthetic datasets to mimic scenarios one might investigate in social science research. We then applied the GATOS workflow to these simulated datasets and compared the themes generated by the workflow to the sub-themes used to generate the original synthetic data. We found the GATOS workflow was able to capture the themes and sub-themes in the synthetic data with a high degree of accuracy. The workflow was able to generate themes that closely matched the original sub-themes in the synthetic data, with only a few instances where the match was not ideal. These findings have implications for humanities and social science research and other fields that rely on qualitative data analysis on large scales. Future research should explore the scalability of the method to larger datasets and test the method on other data with more complex structures. We believe the GATOS workflow has the potential to transform qualitative data analysis and support researchers in generating insights from qualitative data more efficiently and effectively.

Data availability

The simulated data for this study will be made available in the corresponding author's GitHub repository: <https://github.com/andrewskatz>.

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Author contributions

AK wrote the prompts to generate the simulated data; analyzed the data; created the figures; and wrote the introduction, methods, results, discussion, and conclusion sections of the paper. GCF wrote the background section. JBM contributed to editing and conceptualizing. All authors reviewed the full paper.

Competing interests

The authors declare no competing interests.

Ethical approval

Not obtained because the research does not involve human participants or their data. The study used simulated data.

Informed consent

Not obtained because the research does not involve human participants or their data. The study used simulated data.

Additional information

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Correspondence and requests for materials should be addressed to Andrew Katz.

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